

Individual Project Proposal

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ALY6080 Integrated Experiential Learning

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Mar 13, 2026

Background

Cross-border healthcare is booming due to the influx of patients traveling outside of the United States to receive procedures at a fraction of the cost. Increases in procedure costs at home alongside increased wait times and restricted access elective procedures have fueled patients' need for alternative care continuums. Technology platforms are now emerging to bridge patients to international providers and allow organizations to run analytics to better understand cost structures and operations (Batko, 2022).

CareEscapes AI is a platform operating within this space. Initially, the sponsor described the key business issue as demonstrating cross-border care can save patients meaningful costs while operating sustainably. After last week's sponsor meetings and group presentation, it was made clear that financial sustainability is not the main driver for the sponsor. The sponsor was much more engaged when discussing if care destinations for dental tourism were competitive on factors besides cost including safety, accreditation, and trust.

This information greatly shifts the lens through which this proposal will be analyzed.

Statement of the Problem

CareEscapes AI's new problem formulation isn't just if the platform can operate at a profit. Instead, can any destinations abroad be objectively demonstrated as competitors on multiple factors at once when considering dental tourism?

Price is a factor but not the only factor. Past literature surrounding medical tourism has found destination competitiveness to be affected not just by price but also by quality, accreditation, safety, and trust (Sarwar et al., 2012). Policy and regulations also impact the development and growth of cross-border care systems (Virani et al., 2020).

Therefore, the central research question of this project becomes:

How do cost differentials, safety indicators, and accreditation levels jointly influence destination competitiveness in cross-border dental tourism?

If CareEscapes AI does not address this problem with a quantitative framework, they will default to pointing at cost differentials alone. Building trust will be harder and longer-term positioning will suffer.

Project Goal and Objectives

The goal of this project is to develop a multi-dimensional analytical framework that evaluates dental tourism destination competitiveness using real sponsor cost data and publicly available macro indicators.

Specific objectives include:

- Quantify dental cost differentials relative to U.S. pricing
- Integrate safety indicators using Global Peace Index data
- Incorporate accreditation signals using JCI organization counts
- Construct a composite competitiveness score
- Visualize trade-offs between cost savings and safety
- Demonstrate how analytics can support sponsor trust-building efforts

This project intentionally moves beyond simple financial modeling and toward destination validation

Data Sources

This project combines sponsor-provided data with publicly available macroeconomic indicators:

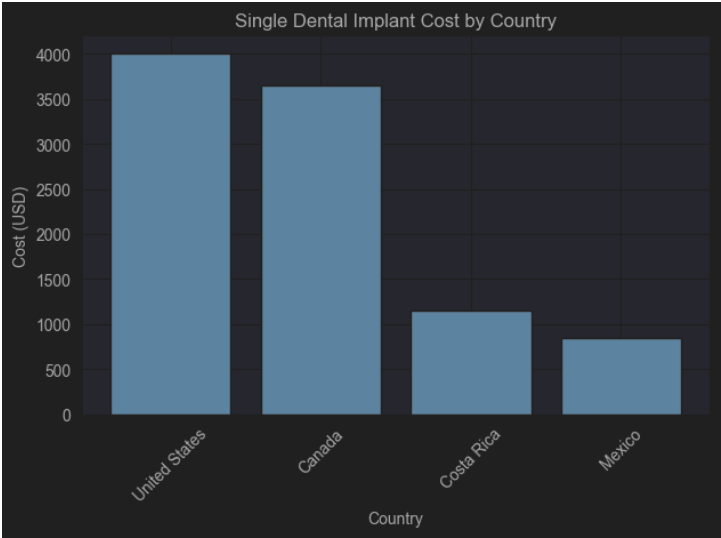
1. Sponsor dental cost data
 - Procedure-level pricing for Single Dental Implant, Abutment & Crown, and Full Arch Restoration
2. Global Peace Index 2025
 - Safety proxy measure
3. World Bank: Current Health Expenditure (% GDP)
 - Healthcare system investment proxy
4. Joint Commission International (JCI)
 - Count of accredited organizations by country

All data are aggregated at the country-procedure level. No patient-level information is used.

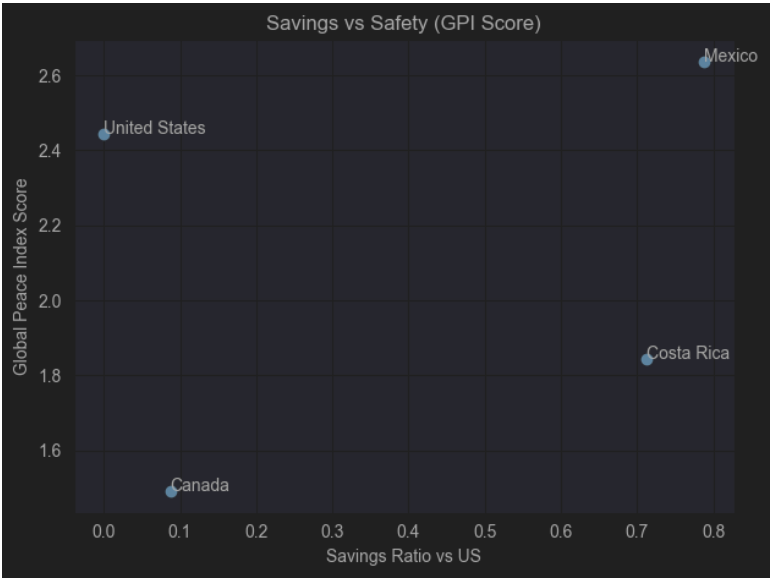
Proposed Methodology

Map of the middle price of one implant dental procedure in select countries calculated from reported price from sponsors. Prices are relative to baseline in United

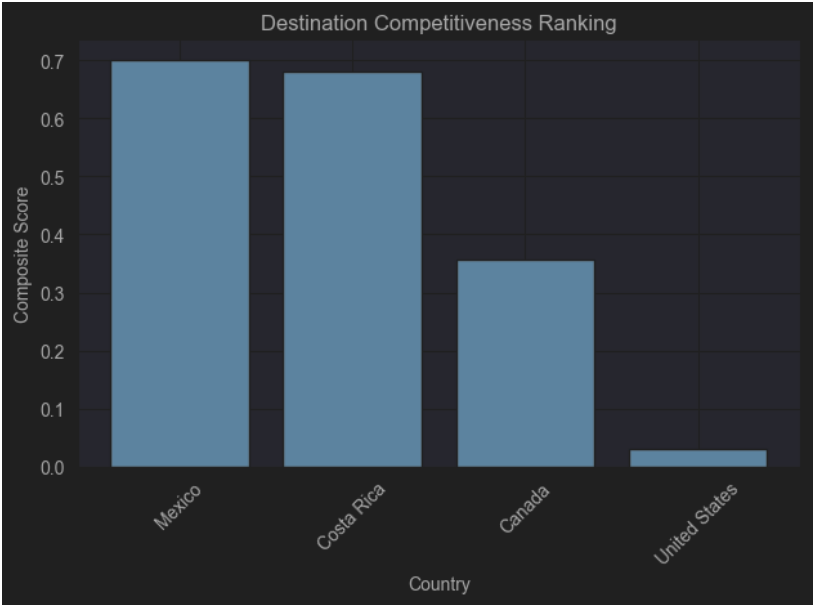
States. Legend shows the difference in price compared to United States.



Scatterplot showing affordability as compared to United States against country safety as measured by Global Peace Index in 2025. Points further to the right are less expensive than the United States and a lower Global Peace Index ranking means the country is safer. This graph shows the affordability and peace of a country in one graph.



This figure presents a composite competitiveness score derived from normalized measures of cost savings, safety indicators, and Joint Commission International accreditation counts. The weighted index was constructed to provide a multi-dimensional assessment of destination attractiveness in cross-border dental tourism.



Exploratory Regression

An exploratory OLS regression is conducted:

Dependent variable: log procedure cost

Independent variables: safety index, healthcare expenditure, accreditation count

This is not interpreted causally. Instead, it identifies statistical relationships that may inform further analysis (Elragal, 2023).

OLS Regression Results									
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Dep. Variable:	log_cost	R-squared:	0.989						
Model:	OLS	Adj. R-squared:	0.980						
Method:	Least Squares	F-statistic:	107.1						
Date:	Sat, 28 Feb 2026	Prob (F-statistic):	8.81e-06						
Time:	19:44:55	Log-Likelihood:	7.3360						
No. Observations:	12	AIC:	-2.672						
Df Residuals:	6	BIC:	0.2375						
Df Model:	5								
Covariance Type:	nonrobust								
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	coef	std err	t	P> t	[0.025	0.975]			

Intercept	7.0135	0.296	23.724	0.000	6.290	7.737			
C(Procedure)[T.Full Arch Restoration (All-on-4 per arch)]	2.5163	0.131	19.165	0.000	2.195	2.838			
C(Procedure)[T.Single Dental Implant]	0.2832	0.131	2.157	0.074	-0.038	0.605			
GPI_2025_Score	-1.0695	0.382	-2.799	0.031	-2.005	-0.134			
Health_Exp_GDP	0.2077	0.050	4.123	0.006	0.084	0.331			
JCI_Count_Approx	0.1340	0.068	1.979	0.095	-0.032	0.300			
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Omnibus:	2.892	Durbin-Watson:	1.670						
Prob(Omnibus):	0.236	Jarque-Bera (JB):	1.028						
Skew:	-0.066	Prob(JB):	0.598						
Kurtosis:	1.572	Cond. No.	89.5						
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Notes:									
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.									

Sponsor Feedback Integration

Originally, the proposal centered on financial sustainability and revenue modeling. However, sponsor feedback during group presentations revealed that they are more interested in validating:

- Whether destinations are safe
- Whether age groups prefer specific procedures
- Whether savings are substantial relative to North America
- How to build trust with potential patients

This proposal responds directly to that feedback by shifting the primary analytical focus toward destination competitiveness rather than internal platform profitability.

Competitor Platform Analysis

An analysis of prominent international medical travel platforms yields little evidence of competition based on price alone. Messaging from competitor platforms overwhelmingly positions value proposition along dimensions of safety, quality control, and coordination support. Provider platform Medical Departures notes that it has processed over 300,000 bookings and partners with over 3,300 doctors at over 2,700 clinics across 37 countries; additionally, the platform lists background checks including credentials verification, clinic accreditation, legal credentials checks, malpractice checks, and professional license checks. Patient reviews, physical site visits, and optional travel insurance support for post-treatment complications are also noted.

Competitor Qunomedical brands itself a patient-centric platform offering 24/7 personal support and access to over 1,000 doctors across over 25 countries. Doctor and clinic listings are all highlighted as having been “vetted by Qunomedical using a unique assessment system called Qunoscore.” The company notes that it works with doctors and clinics on behalf of patients and ensures support throughout the treatment process.

Booking Health stresses its use of “data-driven decision making” when selecting hospitals. The platform claims that its AI-based ranking algorithm assesses over 400 metrics related to quality of care. Visa assistance, interpreters, medical-record translation, transfers, insurance to cover unforeseen medical costs, and even up to one year of supported medical care post-treatment are also noted.

International medical tourism facilitator FlyMedi also concentrates its value proposition messaging around safety and process transparency. FlyMedi promotional

content explains that it only partners with clinics that are “licensed or accredited” and that it has a “rigorous verification process.” End-to-end trip coordination services spanning from consultation to recovery are advertised.

In summation, signals from competitor platforms indicate that the prevailing market standard for medical travel platforms is not defined by “low-cost treatment abroad,” but by “low-cost treatment abroad that is visibly vetted, coordinated, and supported.” This will have ramifications for CareEscapes AI.

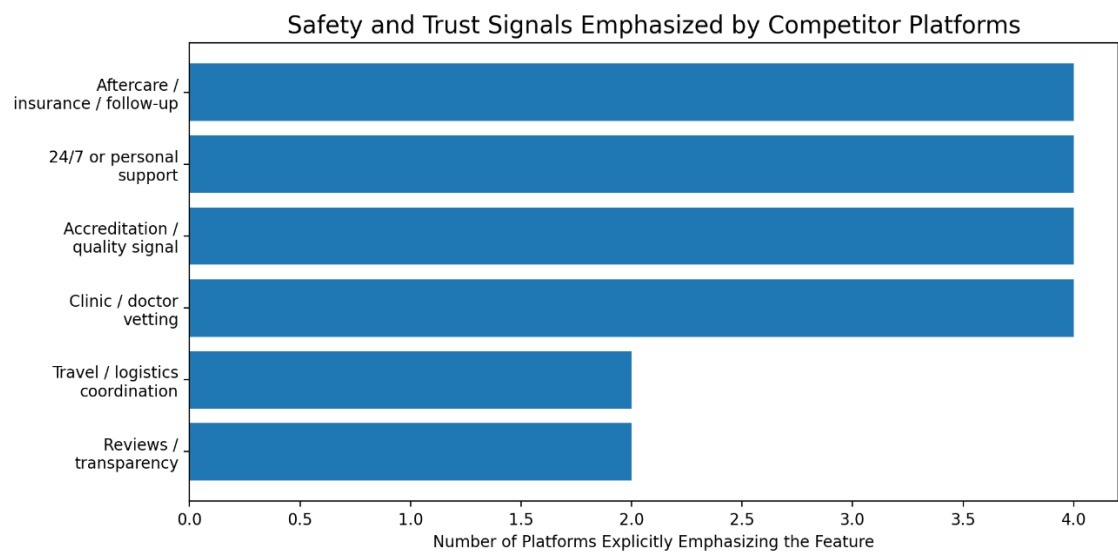
Competitor Platform Comparison: Trust and Safety Signals

	Clinic / doctor vetting	Accreditation / quality signal	Reviews / transparency	24/7 or personal support	Travel / logistics coordination	Aftercare / insurance / follow-up
Medical Departures	✓	✓	✓	✓	—	✓
Quino-medical	✓	✓	—	✓	—	✓
Booking Health	✓	✓	✓	✓	✓	✓
FlyMedi	✓	✓	—	✓	✓	✓

Safety-Focused Sponsor Implication

The sponsor’s number one concern has always been safety. And the competitor review validates this concern. Among the platforms we reviewed, safety was communicated, almost never as a single medical outcome variable. Rather, safety was operationalized by way of a bundle or cluster of trust signals that collectively communicated safety, such as accreditation, physician and clinic verification, transparent reviews from peers who had traveled, coordination of travel if needed, transparency around insurance or covering of complications if they should occur, etc.

Findings here indicate that CareEscapes AI should not limit safety into a risk factor that may vary at the country-level. Safety needs to be considered from the architectural lens of building trust. Applied here would mean that the sponsor may want to layer macro indicators for the safety of a country with micro signals from the platform level like displaying accreditation, verified providers, travel support, follow-up care, etc. Implication for this project is that competitiveness of a destination is tied to both trust and affordability, not just low prices.



Ethical and Practical Considerations

This project does not use identifiable health data. However, cross-border healthcare involves regulatory complexity. Analytics outputs should be interpreted cautiously and used to support managerial decision-making rather than replace it.

Medical tourism growth can influence healthcare systems unevenly (Gholipour & Esfandiar, 2025). Therefore, responsible communication of findings is essential.

Literature Review

Virani, A., Wellstead, A. M., & Howlett, M. (2020). The north-south policy divide in transnational healthcare: A comparative review of policy research on medical tourism in source and destination countries. *Globalization and Health*, 16, 37. <https://doi.org/10.1186/s12992-020-00566-3>

In the comparative bibliometric analysis of policy-relevant medical tourism research related to selected source countries (USA, Canada and the UK) and destination countries (Mexico, India, Thailand, Malaysia and Singapore), the authors find substantial differences between policy priorities and the research agenda in source and destination countries based on a systematic search of the Web of Science database and a review of relevant grey literature. The purpose of this article is to compare the situation of medical tourism in the policy agendas of source and destination countries. The authors analyze policy-relevant research in the field of medical tourism using a systematic search of the Web of Science database. The policy focus of this research is categorized and compared for four source and six destination countries. Grey literature on national policy contexts is reviewed to supplement this comparison. The results show marked differences between source and destination countries in terms of policy issues, with destination country policy and research agendas centered on competitiveness, infrastructure, and costs with limited concern for equity, regulation, and health system impact. In source countries, policy and research concerns are focused on protection of outbound patients, cost containment, quality, and market regulation with limited policy attention to the conditions that drive patient mobility. The article also finds that policy and research on medical tourism remain largely disconnected, with research generally not focused on the issues of highest policy priority and policy decisions made with limited empirical input. The paper concludes that the medical tourism research agenda is still rather disconnected from policy needs, but one promising line of research is a more integrated and comparative policy research agenda.

This article is applicable to CareEscapes AI's core business question as we look to validate the sustainability and operational viability of cross-border healthcare solutions. It is interesting to see, regardless of an outbound patient's motivation or the provider's business case, cost advantage cannot justify cross-border healthcare and its policy environment in isolation. In order to make investment and proposition work, other factors, particularly policy constraints and health system impact in both source and destination countries, need to be accounted for

to ensure a holistic view of the policy and economic situation within which the cross-border healthcare solution is being evaluated. For CareEscapes AI's financial and cost analytics, this research speaks to a similar need to contextualize the insights within the policy context for both source and destination countries, which is a particular consideration given CareEscapes AI's transnational nature of operations and analytics approach to enable data-driven transparency around assumptions related to cost, quality and scalability of cross-border healthcare propositions and market forces within which these solutions are being compared.

Gholipour, H. F., & Esfandiar, K. (2025). Does medical tourism promote growth in healthcare sector? *European Journal of Health Economics*, 26, 233–241.

<https://doi.org/10.1007/s10198-024-01700-3>

This paper examines the impact of medical tourism on healthcare sector growth and performance using annual panel data from 49 developed countries and emerging markets between 2008 and 2022. The authors investigate whether rising medical tourism revenues are associated with measurable growth in a country's healthcare sector. Healthcare sector growth is proxied by the share of gross value added from health and social work relative to total gross value added in the economy. Per capita medical tourism revenue data are obtained from Euromonitor International. To address potential simultaneity issues and capture dynamic adjustments over time, the authors employ both panel Generalized Method of Moments (GMM) estimation and pooled mean group autoregressive distributed lag (PMG/ARDL) models.

Results using both estimation techniques find a robust and positive relationship between medical tourism revenues and healthcare sector growth. Robustness checks including model specification, removal of crisis years, and an alternate performance metric using the World Economic Forum's Global Competitiveness Index for health support these conclusions. Additionally, medical tourism revenues appear to crowd in consumer health expenditure, which is also positively related to healthcare sector growth. In contrast, government expenditure on health is found to be negatively related to health sector growth, suggesting that government expenditure crowds out growth in this sector. Concluding that the authors state that medical tourism revenues can lead to health sector growth and improved competitiveness. Policy prescriptions should be

careful to distribute the benefits equitably to both public and private sectors in each economy.

A core business question for CareEscapes AI is whether cross-border healthcare can be financially sustainable over time. While medical tourism primarily seeks to reduce treatment costs for patients relative to receiving care in the United States, this study underscores the importance of examining how medical tourism revenues scale and influence healthcare systems at a macro level. Increased revenue inflows may enhance healthcare capacity and performance, but doing so requires a deeper understanding of cost structures and public–private sector interactions. Although this paper does not directly address the cost structure of cross-border care relevant to CareEscapes AI, it provides a strong rationale for framing the sponsor’s data collection and analysis around revenue growth, scalability, and cost drivers, rather than focusing on cost savings alone.

Sarwar, A. A. M., Manaf, N. A., & Omar, A. (2012). Medical tourist’s perception in selecting their destination: A global perspective. *Iranian Journal of Public Health*, 41(8), 1–7.

The objective of this article is to present determinants impacting patient destination choice in medical tourism based upon secondary literature review of previous empirical research as well as theoretical articles that occurred between 2000 and 2011. The authors conclude from previous literature that an increase in cost of health care has compelled US and European citizens to travel abroad for medical care. Several factors are found to influence patient destination choice. These factors include cost to receive treatment, service quality and perceived value associated with treatment destination, type/treatment availability, accessibility constraints, and marketing. Research described in article discovered that cost savings was among the highest reason to receive treatment at specific destination, but not independently as service quality, accreditation, treatment availability and marketing were found to greatly influence patient decision making. As concluded by the authors patients attempt to receive treatment that from a country that may be perceived as cheaper while maintaining quality. Article builds upon research conducted by describing several variables that lead to patient perception. These variables create a conceptual model that can be used to show the relationship between the listed variables and patient perception, however further empirical research is required.

This article very closely aligns to CareEscapes AI's key business question. Our sponsor wants to know if our platform providing cross border healthcare will continue to provide sustainable savings while still attracting patients. This article supports the use of sponsor provided data because although CareEscapes AI is analyzing data based off of prices, revenues and costs; patient demand is influenced by other factors as well. This article shows that some of these factors include quality of care, type of treatment availability, and marketing. Understanding destination volume and reasons why can allow CareEscapes AI to better project future revenue streams.

Conclusion

My proposal reframes CareEscapes AI's analytical question from profitability alone to multi-factor destination competitiveness. By combining sponsor cost data with safety, accreditation, and macro indicators, the project builds a structured and defensible validation model.

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